

Kernel Regression

Advanced Methods for Data Analysis (36-402/36-608)

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1 Linear smoothers and kernels

- Recall our basic setup: we are given i.i.d. samples (x_i, y_i) , $i = 1, \dots, n$ from the model

$$y_i = r(x_i) + \epsilon_i, \quad i = 1, \dots, n,$$

and our goal is to estimate r with some function $\hat{r}$. Assume for now that each $x_i \in \mathbb{R}$ (i.e., the predictors are 1-dimensional)

- We talked about consider $\hat{r}$ in the class of linear smoothers, so that

$$\hat{r}(x) = \sum_{i=1}^n w(x, x_i) \cdot y_i \tag{1}$$

for some choice of weights $w(x, x_i)$. Indeed, both linear regression and k -nearest-neighbors are special cases of this

- Here we will examine another important linear smoother, called *kernel smoothing* or *kernel regression*. We start by defining a kernel function $K : \mathbb{R} \rightarrow \mathbb{R}$, satisfying

$$\int K(x) dx = 1, \quad K(x) = K(-x)$$

- Three common examples are the box kernel:

$$K(x) = \begin{cases} 1/2 & \text{if } |x| \leq 1 \\ 0 & \text{otherwise} \end{cases},$$

the Gaussian kernel:

$$K(x) = \frac{1}{\sqrt{2\pi}} \exp(-x^2/2),$$

and the Epanechnikov kernel:

$$K(x) = \begin{cases} 3/4(1 - x^2) & \text{if } |x| \leq 1 \\ 0 & \text{else} \end{cases}$$

- Given a choice of kernel K , and a bandwidth h , kernel regression is defined by taking

$$w(x, x_i) = \frac{K\left(\frac{x_i - x}{h}\right)}{\sum_{j=1}^n K\left(\frac{x_j - x}{h}\right)}$$

in the linear smoother form (1). In other words, the kernel regression estimator is

$$\hat{r}(x) = \frac{\sum_{i=1}^n K\left(\frac{x_i - x}{h}\right) \cdot y_i}{\sum_{i=1}^n K\left(\frac{x_i - x}{h}\right)}$$

- What is this doing? This is a weighted average of y_i values. Think about laying down a Gaussian kernel around a specific query point x , and evaluating its height at each x_i in order to determine the weight associated with y_i
- Because these weights are smoothly varying with x , the kernel regression estimator $\hat{r}(x)$ itself is also smoothly varying with x ; compare this to k -nearest-neighbors regression
- What's in the choice of kernel? Different kernels can give different results. But many of the common kernels tend to produce similar estimators; e.g., Gaussian vs. Epanechnikov, there's not a huge difference
- A much bigger difference comes from choosing different bandwidth values h . What's the tradeoff present when we vary h ? Hint: as we've mentioned before, you should always keep these two quantities in mind ...

2 Bias and variance of kernels

- At a fixed query point x , recall our fundamental decomposition

$$\begin{aligned}\mathbb{E}[\text{TestErr}(\hat{r}(x))] &= \mathbb{E}[(Y - \hat{r}(x))^2 | X = x] \\ &= \sigma^2 + \text{Bias}(\hat{r}(x))^2 + \text{Var}(\hat{r}(x)).\end{aligned}$$

So what is the bias and variance of the kernel regression estimator?

- Fortunately, these can actually roughly be worked out theoretically, under some smoothness assumptions on r (and other assumptions). We can show that

$$\text{Bias}(\hat{r}(x))^2 = (\mathbb{E}[\hat{r}(x)] - r(x))^2 \leq C_1 h^2$$

and

$$\text{Var}(\hat{r}(x)) \leq \frac{C_2}{nh},$$

for some constants C_1 and C_2 . Does this make sense? What happens to the bias and variance as h shrinks? As h grows?

- This means that

$$\mathbb{E}[\text{TestErr}(\hat{r}(x))] = \sigma^2 + C_1 h^2 + \frac{C_2}{nh}.$$

We can find the best bandwidth h , i.e., the one minimizing test error, by differentiating and setting equal to 0: this yields

$$h = \frac{C_2}{2C_1 n^{1/3}}.$$

Is this a realistic choice for the bandwidth? Problem is that we don't know C_1 and C_2 ! (And even if we did, it may not be a good idea to use this ... why?)

3 Practical considerations, multiple dimensions

- In practice, we tend to select h by, you guessed it, cross-validation
- Kernels can actually suffer bad bias at the boundaries ... why? Think of the asymmetry of the weights

- In multiple dimensions, say, each $x_i \in \mathbb{R}^p$, we can easily use kernels, we just replace $x_i - x$ in the kernel argument by $\|x_i - x\|_2$, so that the multivariate kernel regression estimator is

$$\hat{r}(x) = \frac{\sum_{i=1}^n K\left(\frac{\|x_i - x\|_2}{h}\right) \cdot y_i}{\sum_{i=1}^n K\left(\frac{\|x_i - x\|_2}{h}\right)}$$

- The same calculations as those that went into producing the bias and variance bounds above can be done in this multivariate case, showing that

$$\text{Bias}(\hat{r}(x))^2 \leq \tilde{C}_1 h^2$$

and

$$\text{Var}(\hat{r}(x)) \leq \frac{\tilde{C}_2}{nh^p}.$$

Why is the variance so strongly affected now by the dimension p ? What is the optimal h , now?

- A little later we'll see an alternative extension to higher dimensions that doesn't nearly suffer the same variance; this is called an additive model